



Advanced Threat Solution

tentokrát více o koncovém zařízení

Listopad 2018

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Security Strategy Overview

Digital Disruption Drives the Hacker Economy



Attack Surface



Threat Actors



Attack Sophistication

...Creating an ever-evolving, dynamic threat landscape

- 
- A world map with a network overlay of dots and lines, representing global connectivity. The map is dark blue, and the network lines are a lighter blue. The text is white, and the red arrowheads are a bright red color.
- ▶ Ransomware
 - ▶ DDoS
 - ▶ Advanced Persistent Threats
 - ▶ Malvertising
 - ▶ Drive by Downloads
 - ▶ Data Destruction
 - ▶ Data Manipulation
 - ▶ Spyware/Malware
 - ▶ Data/IP Theft
 - ▶ Phishing
 - ▶ Monetary Theft
 - ▶ Unpatched Software
 - ▶ Wiper Attacks
 - ▶ Rogue Software
 - ▶ Botnets
 - ▶ Man in the Middle
 - ▶ Trojans

Visibility Across Threat Vectors

You Can't Find What You Can't See



Endpoint



Branch



Edge



Campus



Data
Center



Cloud



Operational
Technology

We developed Cisco Talos: the largest non-government threat intelligence organization on the planet



250+ full-time threat
researchers and data
scientists



Analyzing 1.5 million
unique malware
samples daily



Blocking 20 billion
threats daily. More than
20x any other vendor.

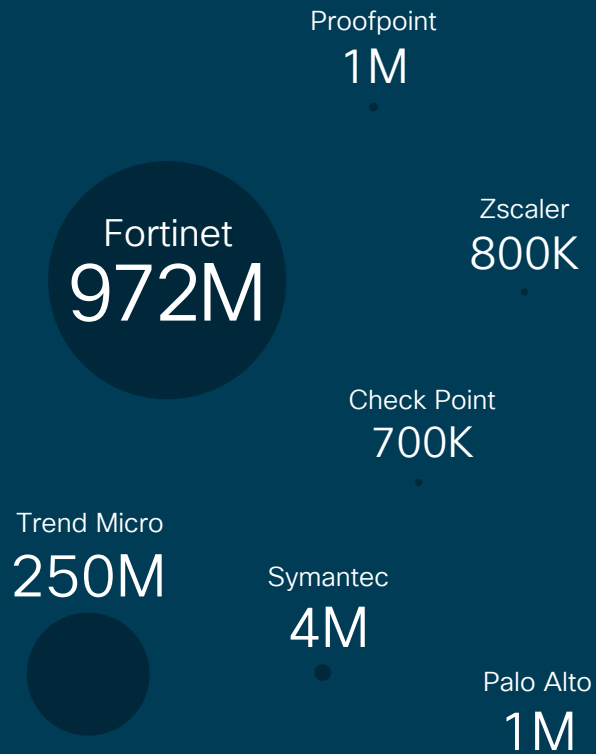
We **see more** so you can **block more** and **respond faster** to threats.

More threats blocked
daily than anyone else

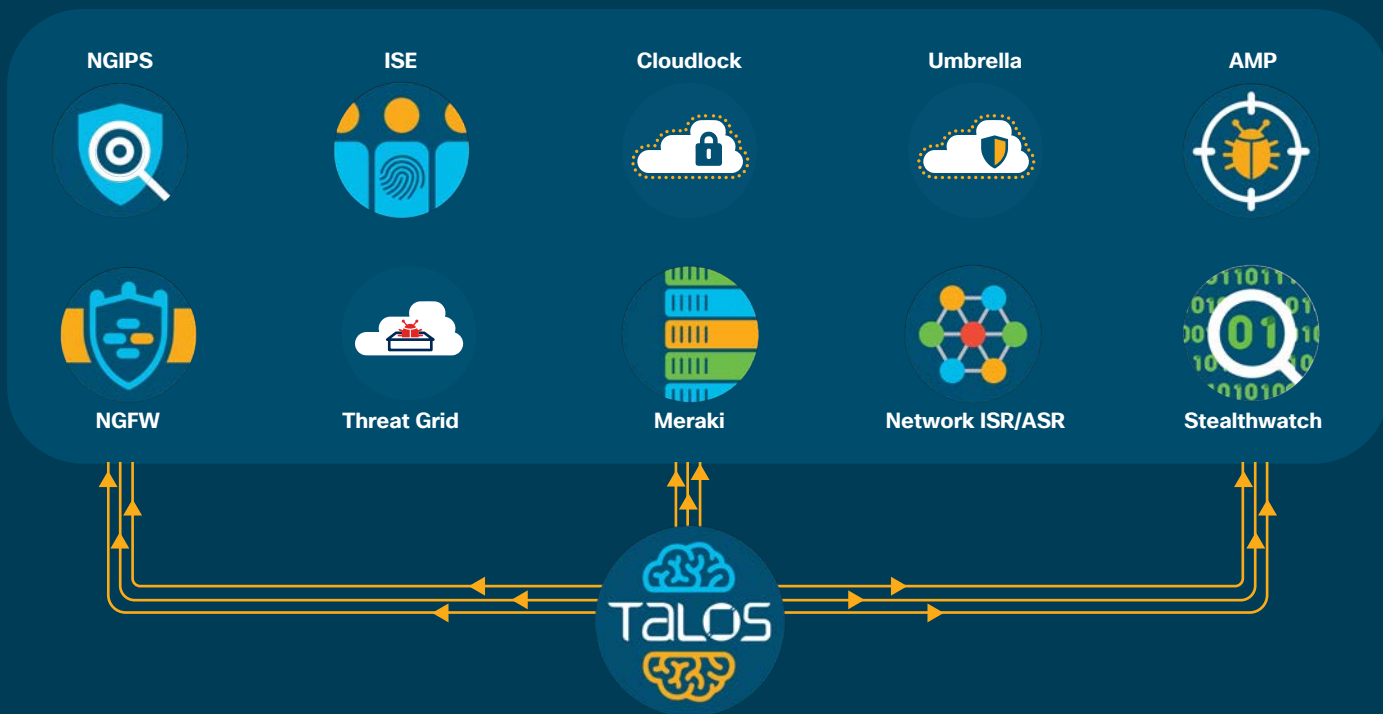
TALOS

Cisco Security Research

20B



See it once, protect everywhere



Best news yet: Cisco Talos is free for customers

Forcing the Bad Guys to Innovate



Spreading security news,
updates, and other
information to the public



BEERS



TALOS

Podcast



Download on
iTunes



GET IT ON
Google Play

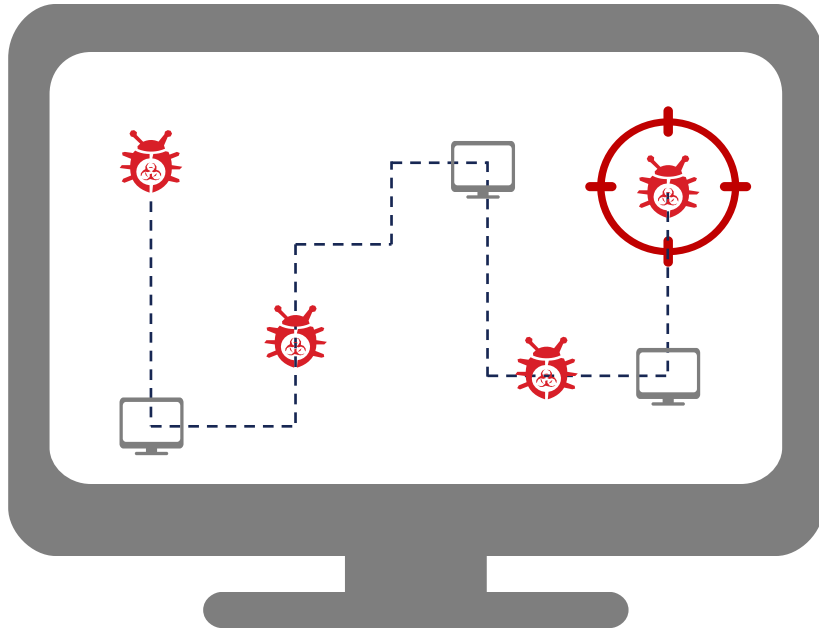
TALOS

AMP for Endpoints

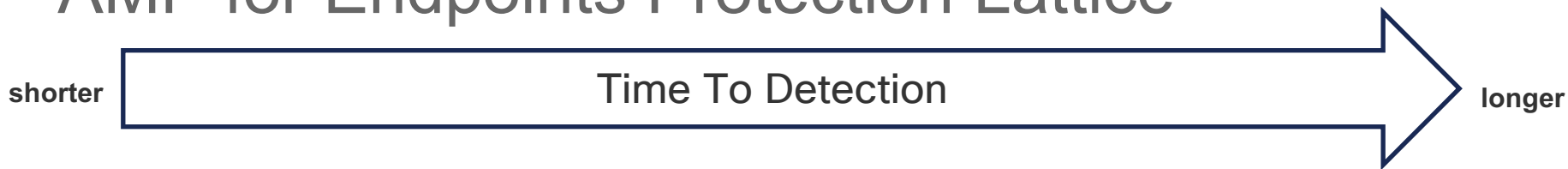
Continuous Analysis and Retrospective Security



● Recording



AMP for Endpoints Protection Lattice



In Memory

Exploit Prevention

System Process Protection

On Disk

AMP Cloud

Malicious Activity Protection

TETRA

Custom Detections

Post-Infection

Cognitive Threat Analytics

Device Flow Correlation

Cloud IOCs

Endpoint IOCs

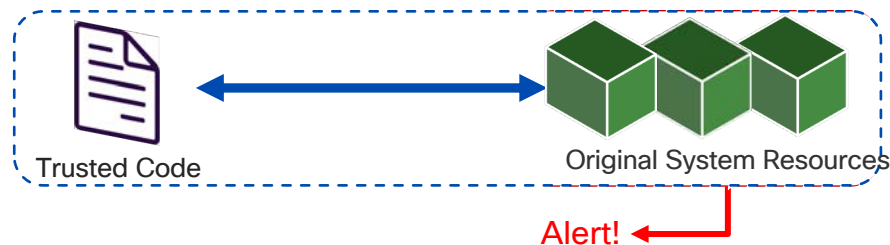
Exploit Prevention

- Make the memory unpredictable by changing its structure
- Make the app aware of legitimate memory structure
- Any code accessing the old structure is malware
- Currently protects 32-bit apps on 32/64-bit OS! (64bit app protection coming in AMP for Windows 6.2.x, check release notes)

- excel.exe
- winword.exe
- powerpnt.exe
- outlook.exe
- iexplore.exe

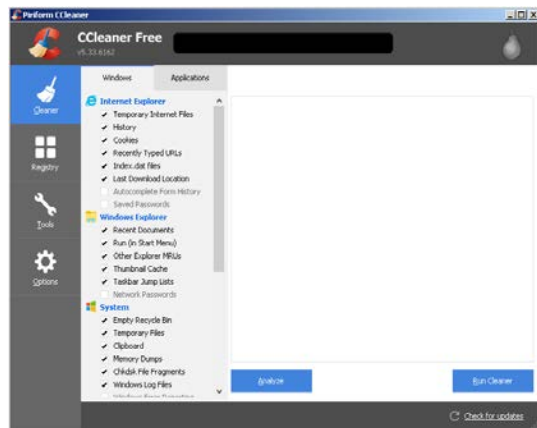
- firefox.exe
- chrome.exe
- skype.exe
- teamviewer.exe
- vlc.exe

- wscript.exe
- powershell.exe
- acro32.exe
- rundll.exe
- taskeng.exe



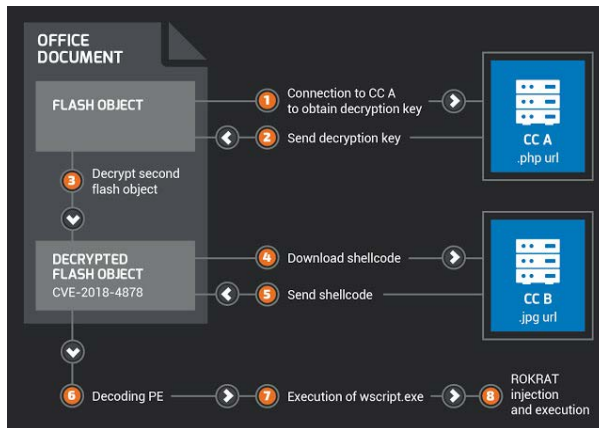
Exploit Prevention: In Field Findings

CCleaner



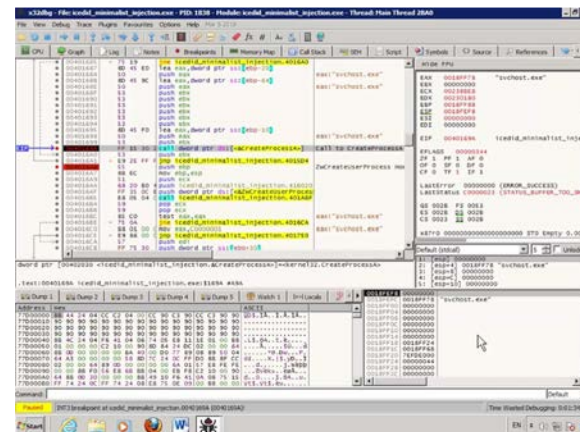
ExPrev Beta Test leads to backdoor discovery in CCleaner software from Avast

0-day Flash



0-day Remote Code Execution vulnerability prevented, prevents exfiltration and remote admin

IcedID Trojan



Minimalist (evolutionary) code injection technique prevented by ExPrev technology

CCleaner: <https://blog.talosintelligence.com/2017/09/avast-distributes-malware.html>

0-day Flash: <http://blog.talosintelligence.com/2018/02/group-123-goes-wild.html>

IcedID: Talos Analysis: <https://blog.talosintelligence.com/2018/04/icedid-banking-trojan.html>

System Process Protection

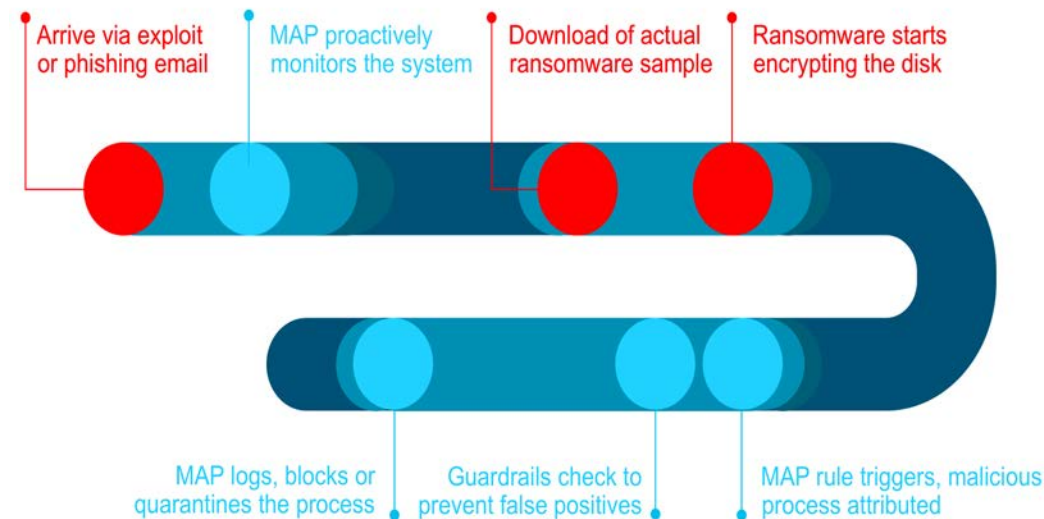
- Protects Windows system processes from being compromised through memory injection attacks
- Evaluates desired process/thread access, truncates potentially dangerous access
- Protects against Mimikatz dumping credentials from lsass.exe memory

- Session Manager Subsystem (smss.exe)
- Client/Server Runtime Subsystem (csrss.exe)
- Local Security Authority Subsystem (lsass.exe)
- Windows Logon Application (winlogon.exe)
- Windows Start-up Application (wininit.exe)

```
.#####.  mimikatz 2.0 alpha (x86) release "Kiwi en C" (Apr  6 2014 22:02:03)
.## ^ ##.
## / \ ## /* * *
## \ / ## Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
'## v ##' http://blog.gentilkiwi.com/mimikatz                (oe.eo)
'#####'                                     with 13 modules * * */
```

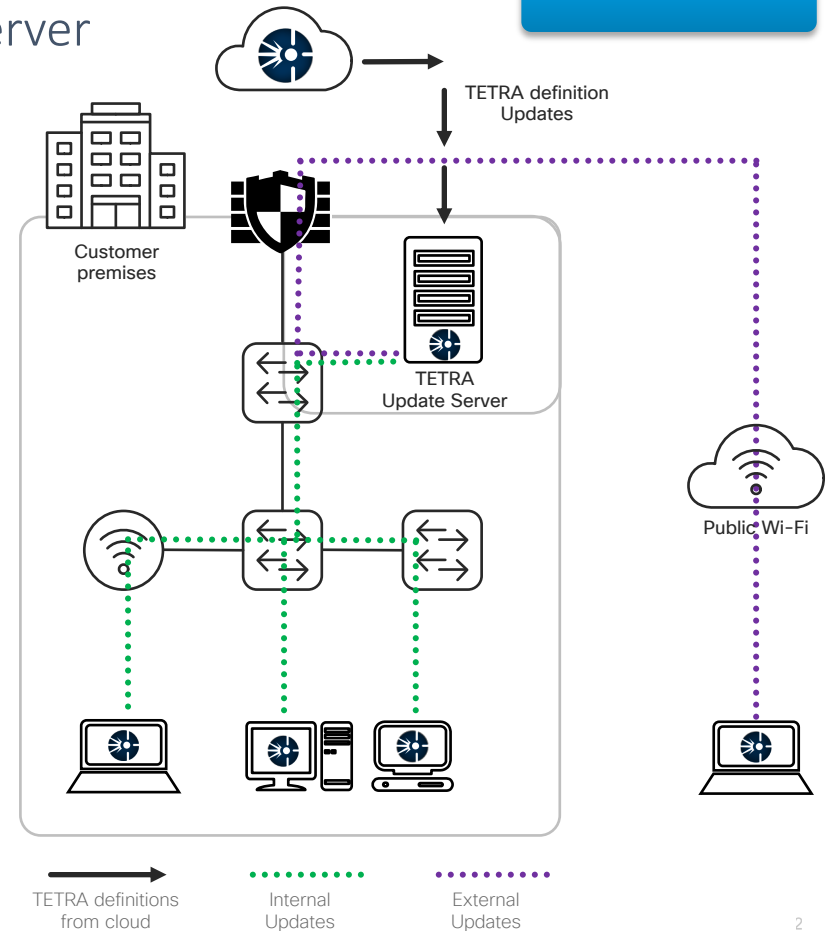

Malicious Activity Protection

- Detects abnormal behavior of a running program, initially focused on ransomware
- Uses rules that monitor processes reading, writing, and renaming or deleting files within short span of time
- Modes of operation: audit, blocking, quarantine
- Process can be excluded from MAP inspection



TETRA and AMP can use also local Update Server

- Offline AV engine for Windows
- On-prem server gets updates from AMP Public Cloud
- Server FQDN configured per AMP Policy
- Can make FQDN available publically for external updates
- AMP Update Server runs on Windows or Linux, uses IIS / Apache / nginx (currently TETRA updates only)

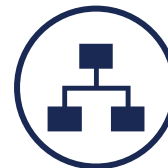


Cognitive Intelligence

- Visibility into devices with or without AMP Connector – cover unsupported OS and IoT devices
- File-less malware and ~30% more detections
- Correlation with AMP for Endpoints events and links to files responsible for C2 communication
- Priority rating and human readable threat descriptions with course of action



Data
Exfiltration



C&C
Communication



DGAs



Exploit Kits



HTTP(S)
Tunneling

Cisco/Apple Alliance – Protect iOS Users with Cisco Security Connector



New MDM/EMM vendor support:



IBM Security



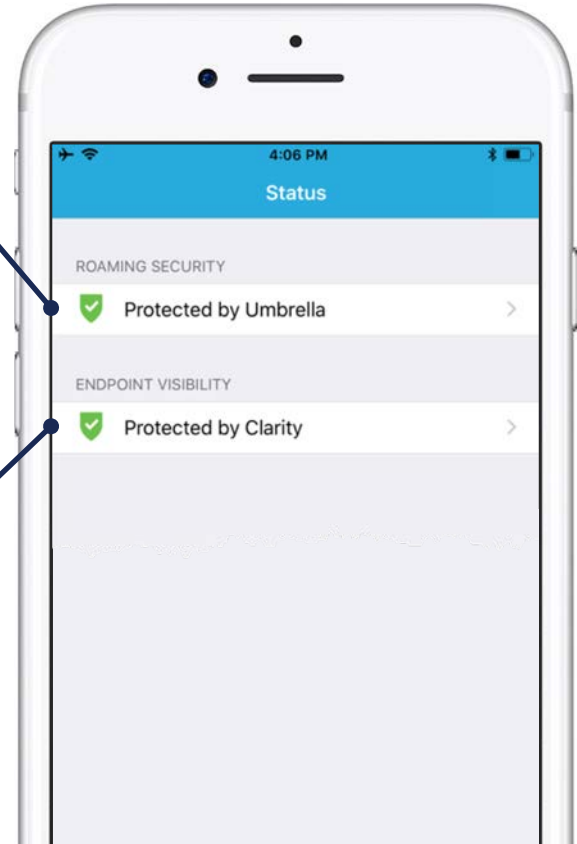
jamf

Control and visibility

- DNS-layer enforcement and encryption via net new iOS 11 functionality
- Customizable URL-based protection with intelligent proxy
- Available to Umbrella* customers at no extra charge if subscription already covers iOS users

Visibility

- App-layer auditing and correlation via net new iOS 11 functionality
- Logs encrypted URL requests without SSL decryption
- Available to AMP for Endpoints customers at no extra charge if subscription already covers iOS devices



* Professional, Insights and Platform packages

Demo

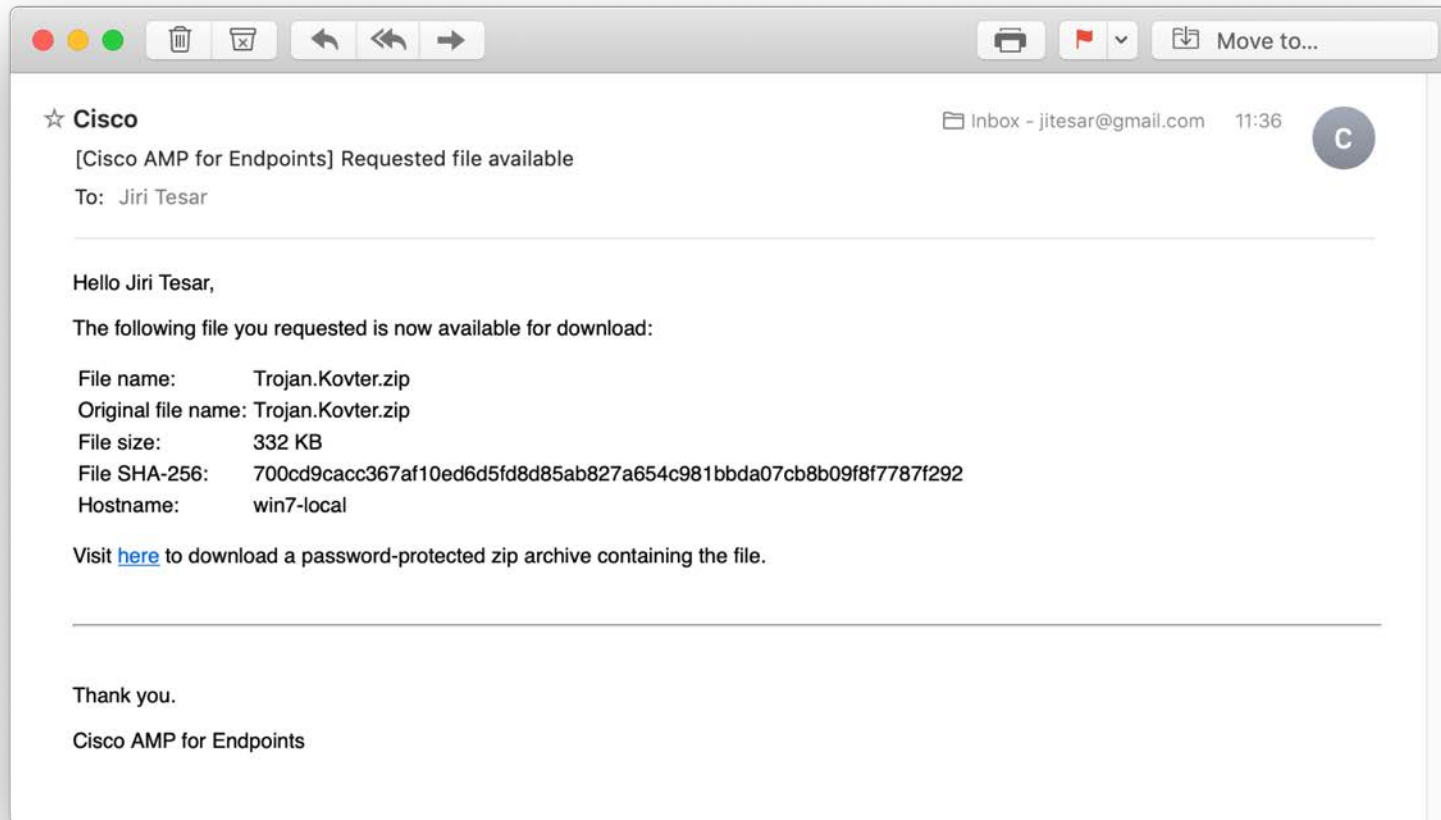
AMP4E

Threatgrid

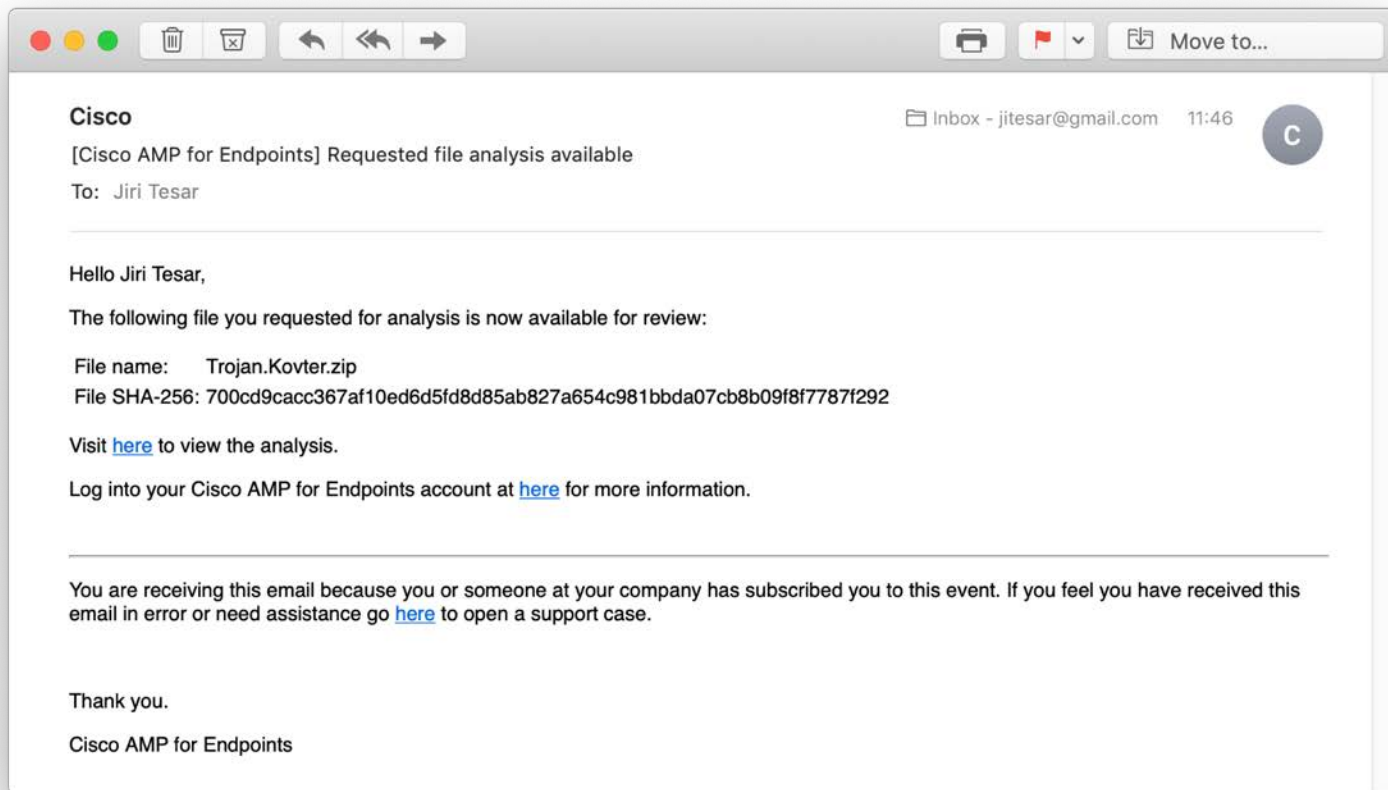
Umbrella

Cisco Threat Response

AMP4E – Fetch the File for Analysis



AMP4E – Fetch the File and Send to Sandbox



Threat Grid

Threat Grid

1. Sample submission

2. Analyze, Correlate, and Enhance

3. Produce Intelligence & Inform AMP Architecture



Input

Submit suspicious samples to Threat Grid via Integration, API, or Portal

Process

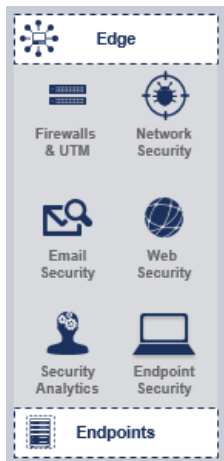
Sample is executed and analyzed using multiple techniques

- Proprietary techniques for static and dynamic analysis
- “Outside looking in” approach
- 1000+ Behavioral Indicators

Output

- Behavioral Indicators & Threat Score
- Pokes AMP cloud, integrations will block
- Threat Intel Feeds & Global Intel

Threat Grid Integrations



Select Recipe Integrations



Select Threat Feed Integrations



File Analysis: Static and Dynamic

- Static Analysis

- File on disc
 - Header details
 - AV engines
- What it **is/contains**



- Dynamic Analysis

- Execution/Detonation
 - Network Connections
 - File/System changes
 - Function/Library calls
- What it **does**



Addressing the Challenges: Playbooks



User Emulation



Automation



- 9 Default Playbooks
- User Generated Playbooks
- Dynamic Playbook Selection



-  Virtual Exit Node
-  TG Cloud
-  TG Appliance
-  C&C Server

Network, Web, Email Security Integrated File Analysis – On Premise Option

**AMP for Networks
IDS / IPS**



**AMP on Web
Security Appliance**



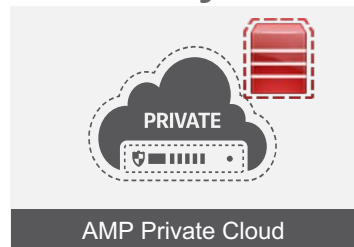
**AMP on Email
Security Appliance**



**AMP Endpoint
Agents**



**AMP File
Analysis**



**AMP ThreatGrid
Sandbox**



**Process names
Registry Keys
IP Addresses
DNS Names**

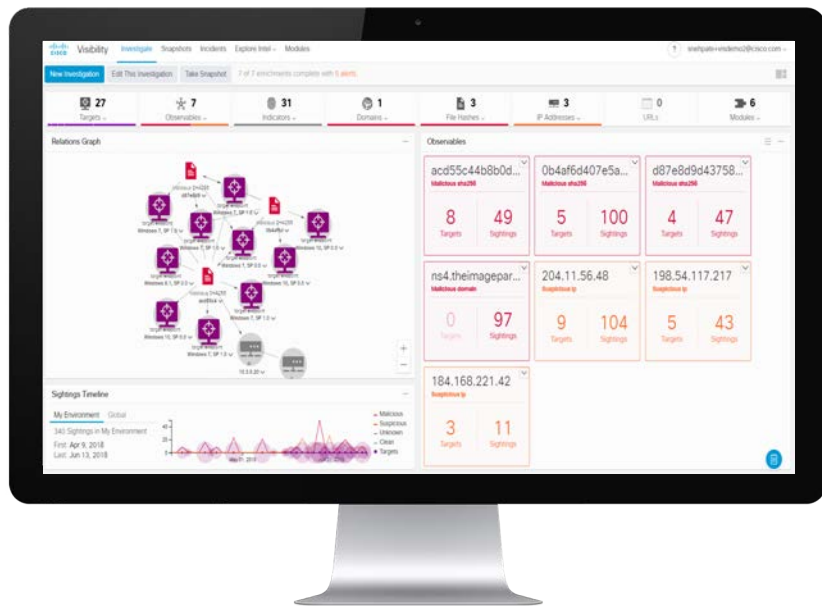


splunk

Cisco Threat Reponse

Cisco Threat Response

Integrating security for faster defense



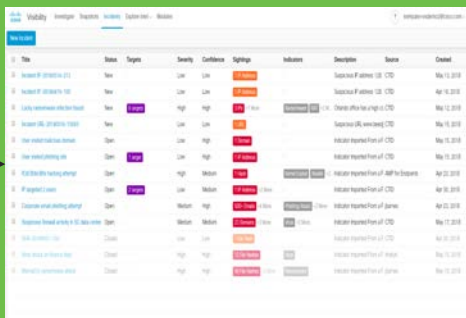
Key pillars of our integrated architecture

- Automates & Orchestrates across security products
- Focuses on security operations functions – Detection, Investigation, and Remediation

Contextual Analysis and Incident Response (support will come also with NFWG and Content Gateways)

Cisco Threat Response

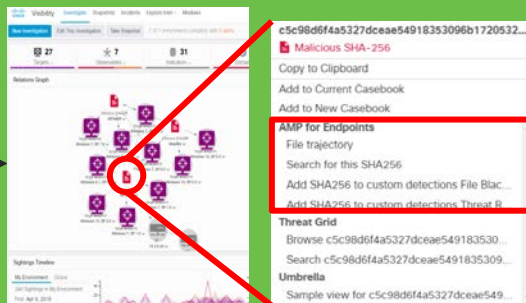
1 Get high fidelity IPS events



2 Investigate with automated enrichment



3 Remediate in AMP & Umbrella



FMC

NFWG

Virus
Total

TALOS

ThreatGrid

AMP

Umbrella

SMA

- From FMC, pivot into Threat Response via casebook browser plug-in

- What do you know about these (IP, Hash, URL, etc.) observables?

- Have we seen these observables?
- Which end-points reached out to the URL?
- Etc.

Encrypted Traffic?

Encrypted Traffic

https inspection on gateways (resign, known keys)

- NGFW
- WSA

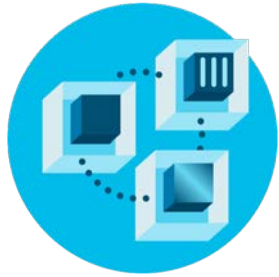
Leverage Endpoint Visibility

- AMP4E
- NVM AnyConnect

Behavior analysis of encrypted traffic

- ETA + Stealthwatch

Encrypted Traffic Analytics (ETA)



Telemetry sources that instrument the digital business.



Collect and store at scale.



Analyze and automate.



Security Outcomes

Catalyst
9000



Stealthwatch
Enterprise



Stealthwatch
Custom Security
Event



Cryptographic
Audit



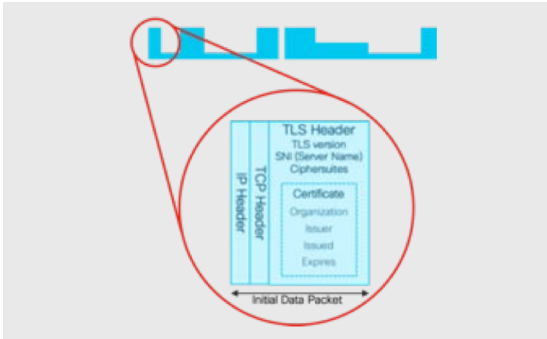
Cognitive
Intelligence



Malware
Detection

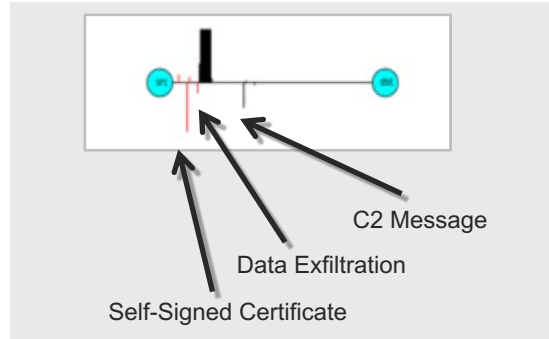
ETA data

Initial Data Packet



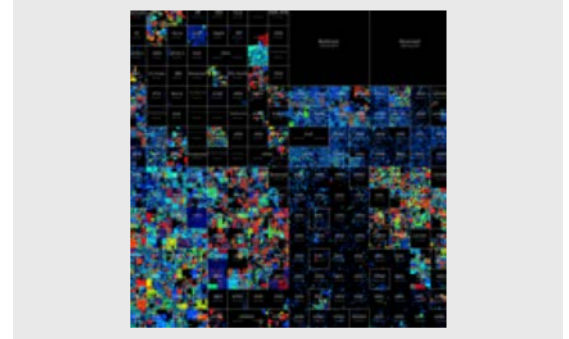
Make the most of unencrypted fields

Sequence of packet lengths and times



Identify the content type through the size and timing of packets

Global Risk Map



Know who's who of the Internet's dark side

Cryptographic Compliance

 **Stealthwatch**

[Dashboards](#) [Monitor](#) [Analyze](#) [Jobs](#)

[Search](#) [User](#)

[Desktop Client](#)

Flow Search Results (143)

[Save Search](#) [Save Results](#) [Start New Search](#)

[Edit Search](#) [Time Range: Last Hour](#)

[100% Complete](#) [Delete Search](#)

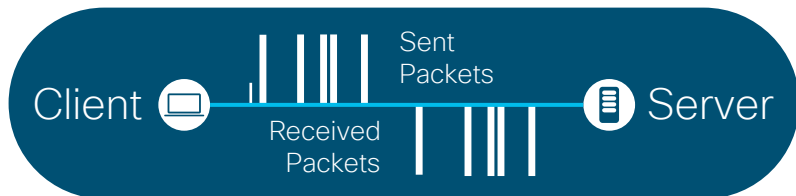
Subject: [10.201.3.51](#) [Orientation: Either](#)

Peer: [Port / Protocol: 443/tcp](#)

START	DURATION	SUBJECT IP ADDRESS	SUBJECT HOST GROUPS	SUBJECT BYTES	CONNECTION BYTES	ENCRYPTION TLS/SSL VERSION	ENCRYPTION KEY EXCHANGE	ENCRYPTION ALGORITHM AND KEY LENGTH	ENCRYPTION AUTHENTICATION ALGORITHM	ENCRYPTION MAC	PEER IP ADDRESS	PEER HOST GROUPS	PEER BYTES
▶ Oct 18, 2017 1:53:02 PM	54s	10.201.3.51 View URL Data	End User Devices,Desktops,Atlanta,Sales and Marketing	125.67K	14.21M	TLS 1.2	ECDHE	AES_128_GCM/128	RSA	SHA256	104.16.41.2	United States	14.09M
▶ Oct 18, 2017 2:34:08 PM	1m 1s	10.201.3.51 View URL Data	End User Devices,Desktops,Atlanta,Sales and Marketing	112.83K	3.67M	TLS 1.2	ECDHE	AES_128_GCM/128	ECDSA	SHA256	31.13.77.36	United States	3.56M

Identifying malicious encrypted traffic

Model

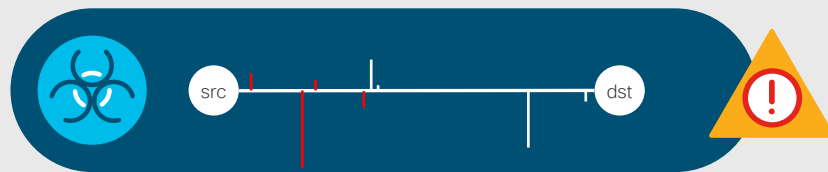


Packet lengths, arrival times and durations tend to be inherently different for malware than benign traffic

Google Search Page Download



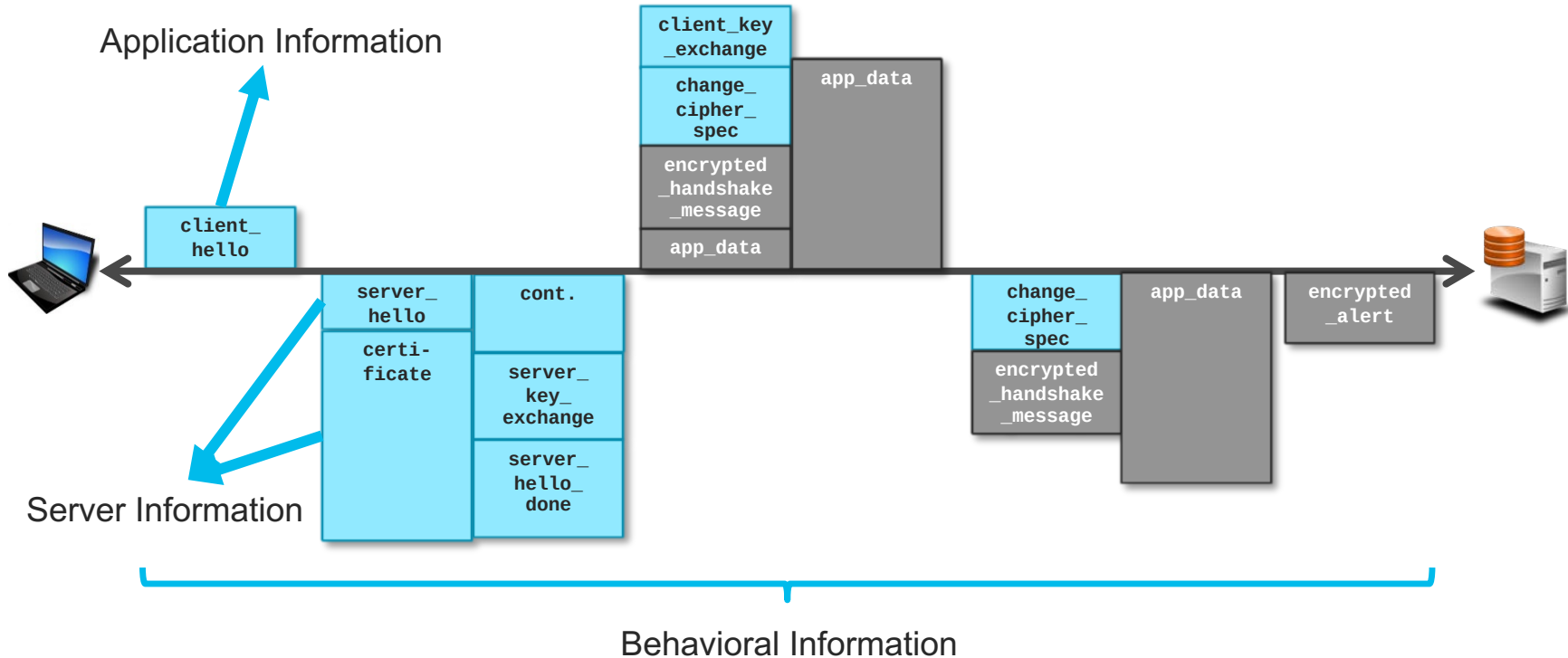
Initiate Command and Control



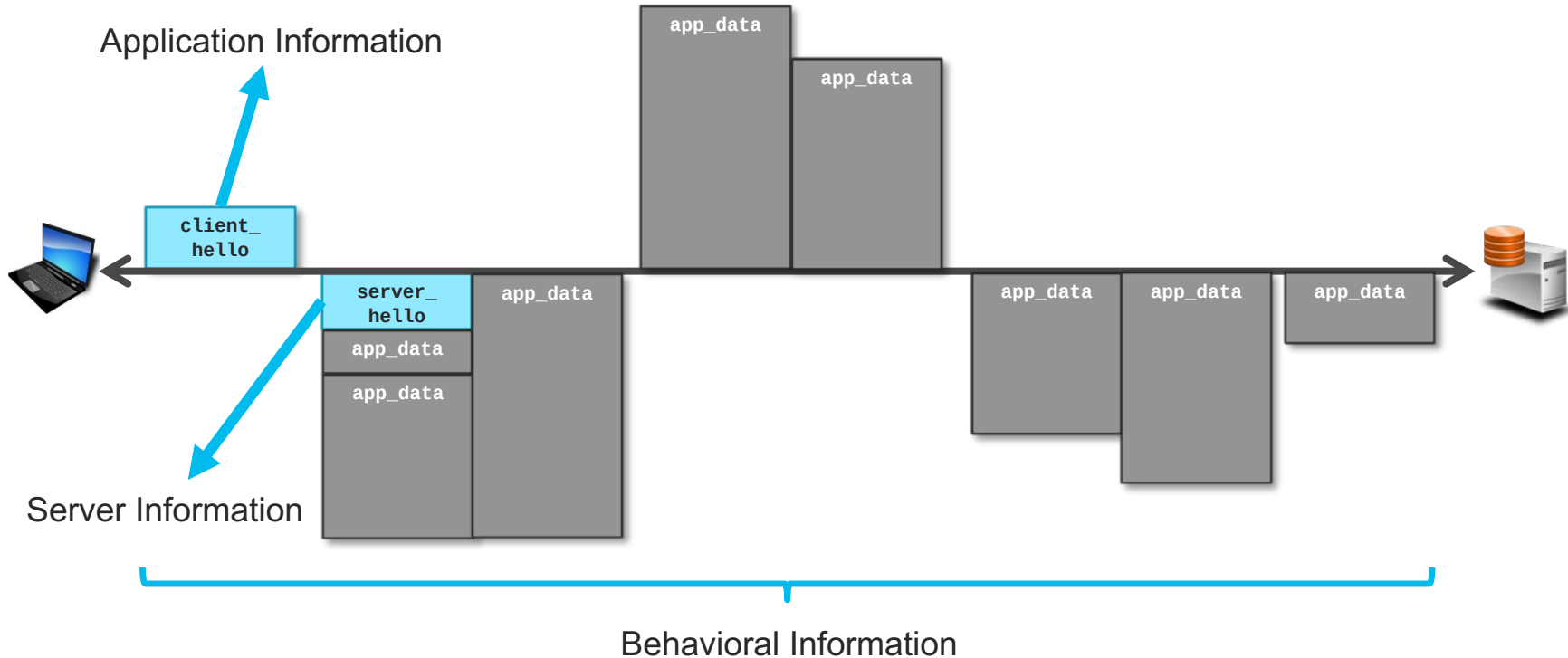
Exfiltration and Keylogging



ETA Data Features, <= TLS 1.2



ETA Data Features, TLS 1.3





- 

- ❖ ~400 Detections using ETA
- ❖ Ransomware detected
- ❖ C&C and Data Exfiltration
- ❖ Multiple Critical, High- and Medium-risk Detections
- ❖ Numerous Malware Instances including Cryptomining & Botnet activities
- ❖ Several Applications using TLS 1.0

The diagram illustrates the network architecture for generating and distributing an ETA Kit. A Management (Mgmt) cloud connects to an ASR1K router (ASR1K) within the ETA Kit. The ASR1K router is connected to a CL Net via a SPAN port and to an Internet Edge ASR1K router. The Internet Edge ASR1K router connects to the Internet cloud, which then connects to a Cognitive Analytics cloud. The ASR1K router also connects to a Service Management Center (SMC) via a secure connection (HTTPS: SCP).

Security Insight Dashboard Inside Hosts

Alarming Hosts

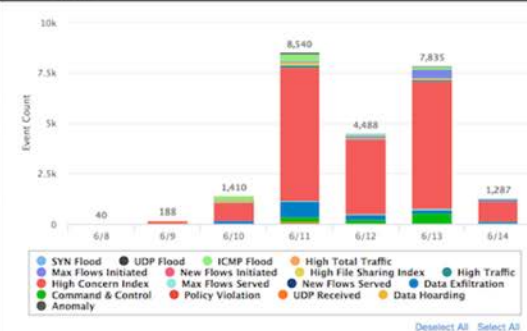


Top Alarming Hosts

HOST	CATEGORY
10.17.20	CI EX
320-Session-Capture	
10.17.20	CI EX
320-Session-Capture	
10.34.0	EX CI
Hyatt Guest Wired	
10.21.178	CI
OCOC-01-02 Guest Wifi	
10.16.0	EX CI
OCOC-01-02 NOC Wireless	
10.55.0	CI EX
WOS Booths	
10.16.70	CI EX
OCOC-01-02 Location	

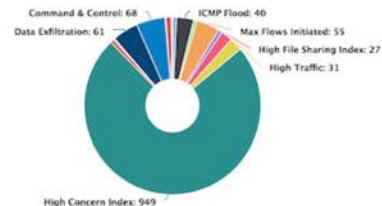
[View All Hosts](#)

Alarms by Type



[Deselect All](#) [Select All](#)

Today's Alarms



Cognitive Threat Analytics

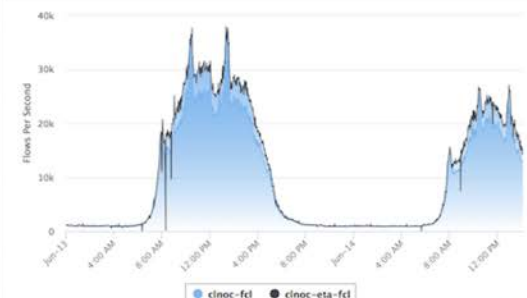
AFFECTED USERS BY RISK

Critical	High	Medium	Low	Total
2	5	65	319	391

10	10.21.131	ENCRYPTED
10	10.21.142	ENCRYPTED
8	10.21.142	ENCRYPTED
8	10.21.208	ENCRYPTED
8	10.21.159	ENCRYPTED
8	10.21.205	ENCRYPTED

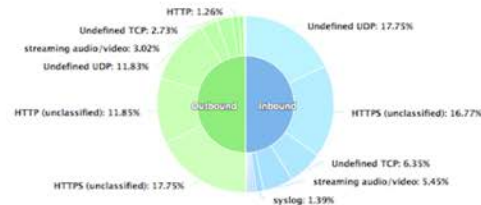
[View Dashboard](#)

Flow Collection Trend

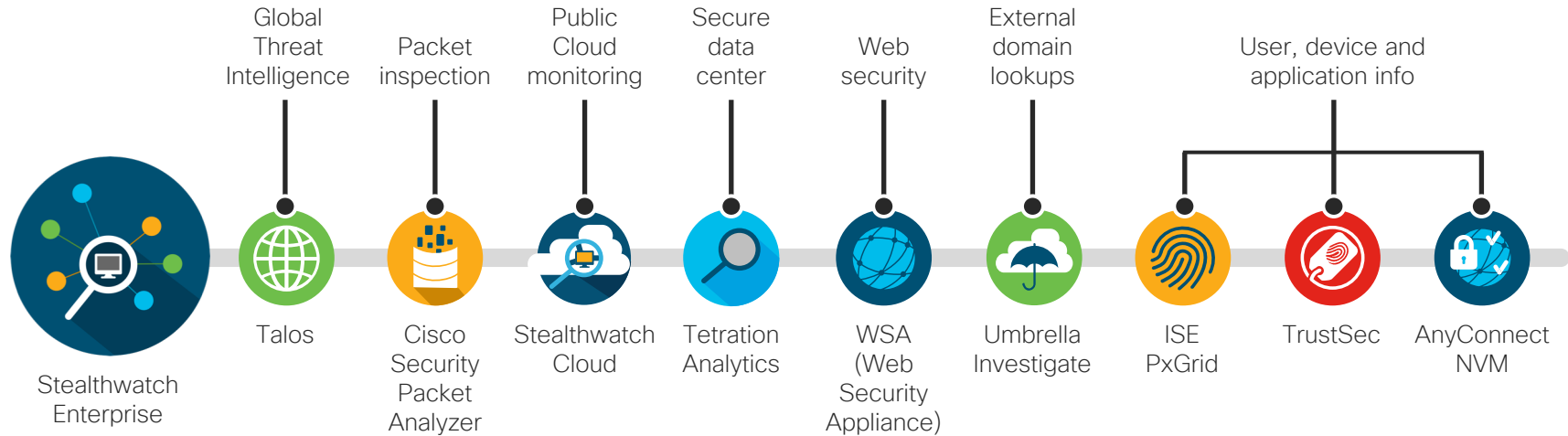


[Deselect All](#) [Select All](#)

Top Applications



Security that works together



Stealthwatch Endpoint Visibility Solution

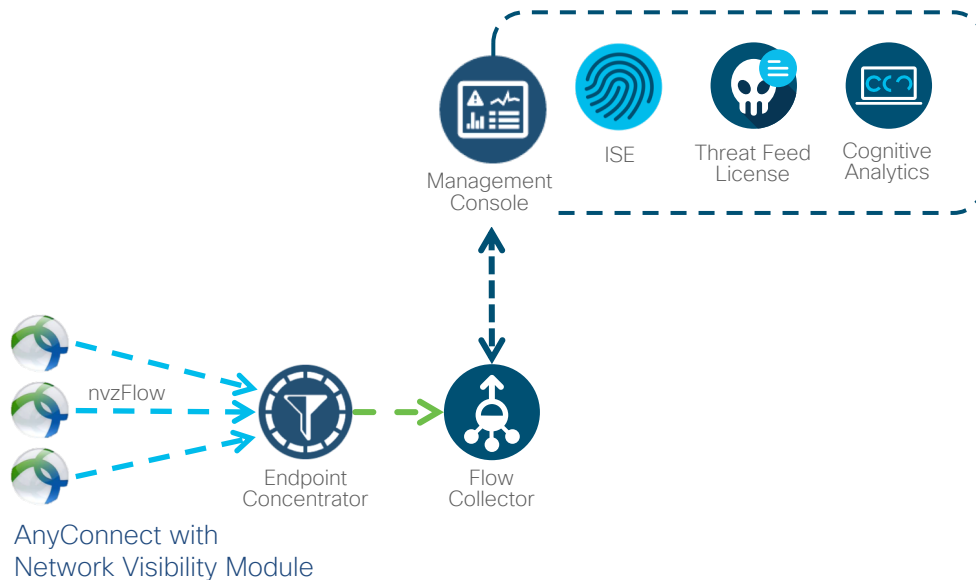
Flow Detailed Summary: 10.100.104.2

Search Subject Details	Totals	Peer Details
Packets: 1.25K Packet Rate: 416pps Bytes: 1.74MB Byte Rate: 607.51Kbps Percent Transfer: 100% Host Groups: Catch All TrustSec Name: Group 1 TrustSec SGT: 27	Packets: 1.25K Packet Rate: 416pps Bytes: 1.74MB Byte Rate: 607.51Kbps Search Subject/Peer Ratio: all search subject RTT: 0s SRT: 0s	Packets: 0 Packet Rate: 0pps Bytes: 0B Byte Rate: 0bps Percent Transfer: 0% Host Groups: Catch All TrustSec Name: Group 2 TrustSec SGT: 52

Process name: malware.exe
File SHA Hash: 6ca13d52ca70c883e0f0bb101e425a89e8624de51db2d2392593af6a84118090

Attributing a flow to:

- Process name
- Process hash
- Process account
- Parent process name
- Parent process hash
- Parent process account



Integrated Security

Latest Announcements in the Cisco Security Technical Alliances Ecosystem

- See 9/19 announcement: [How Alliances Strengthen Your Cybersecurity Defenses](#)
- Introducing pxGrid 2.0 – evolving the bedrock of our policy ecosystem
- ISE & IoT – bringing IoT into mainstream network access policy
- New integrations and partners from network to endpoint to cloud...
 - ISE
 - Firepower
 - AMP for Endpoints
 - Cisco Cloud Security
 - Threat Grid
 - Cisco Security Connector

Appendix – Details on New Ecosystem Partners and Integrations

CSTA September Announcement Summary

57 new integrations from network to endpoint to cloud...



CISCO ISE

*pxGrid Integrations for
IOT, Orchestration,
Deception, Endpoint,
Vulnerability
Management*



CISCO Firepower

*Threat Intelligence Director for
NGFW Enrichment, Firepower
integrations*



CSTA September Announcement Summary

57 new integrations from network to endpoint to cloud...



CISCO AMP for Endpoints

Integrations provide analysts with detailed information and actions on endpoint events

CYBERSPONSE
ADAPTIVE SECURITY

exabeam

IBM Security

LogRhythm

SIEMPLIFY

SYNCURITY



CISCO Cloud Security

Threat Intelligence on Malicious Domains and Threat Response Enforcement & CASB

CYBERSPONSE
ADAPTIVE SECURITY

efficient iP[™]
DEFINING SMART DDI

IBM Security

Menlo
Security

RSA

RAPID7

SYNCURITY



CISCO Threat Grid

Malware Intelligence Sharing and Incident Response Integration

CYBERSPONSE
ADAPTIVE SECURITY

DEMISTO

exabeam

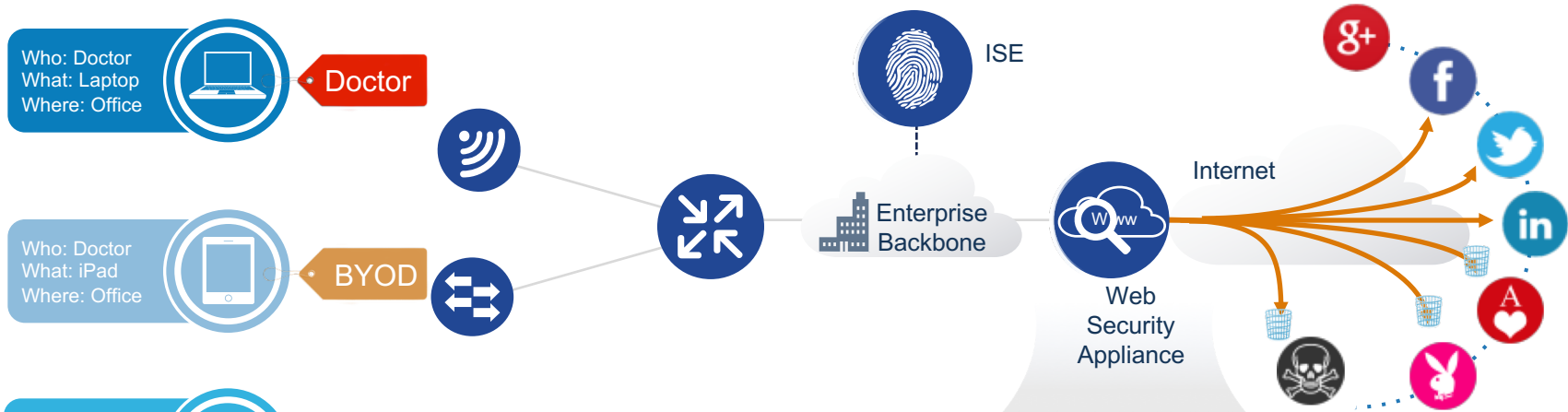
IBM Security

SIEMPLIFY

SYNCURITY



Simplifying WSA Policies with SGTs



Policies						
Order	Group	Protocols and User Agents	URL Filtering	Applications	Objects	Anti-Malware and Reputation
1	Doctors	(global policy)	Block: 1 Monitor: 78	Block: 10 Monitor: 367	(global policy)	(global policy)
2	Doctors BYOD	(global policy)	Block: 1 Monitor: 78	Block: 10 Monitor: 367	(global policy)	(global policy)
3	Guests	(global policy)	Block: 1 Monitor: 78	Block: 10 Monitor: 367	(global policy)	(global policy)
	Global Policies	No blocked items	Monitor: 79	Monitor: 367	No Blocked Items	Web Reputation: Enabled Anti-Malware Scanning: Enabled

ISE as a source of Context

Updated Jan 11, 2018 10:52:08.975 PM Started Show CoA Actions

Endpoint ID	Identity	IP Address	Endpoint Profile	NAS IP Address	Device Port
CC:16:7E:E1:FF:24	adam	10.10.10.138	Windows7-Workstation	172.16.15.1	GigabitEthernet1/0/3

Cisco ISE

- Live Sessions Table of ISE
- Device/User Authentication
- Device Profiling
- NAD details

Identity and Device Table

Filter Domain : MANGO Cisco ISE : ise.mango.local

Identity and Device Table - 2 records

Start Active Time	End Active Time	User Name	Host	MAC Address	Device Type	Network Access Device	Network Access Interface
Jan 11, 2018 10:52:08 PM (9 minutes 12s ago)	Current	adam	10.10.10.138	cc:16:7e:e1:ff:24 (Cisco Systems, Inc)	Windows7-Workstation	distswitch-mgmt-vlan1615.mango.local (172.16.15.1)	GigabitEthernet1/0/3

SMC

- Live Authentication Events shown in SMC

Identity and Device Table adam

Filter Domain : MANGO Time : Today User : adam

Identification Alarms Most Recent Flow

Flow Table (Last 5 minutes) - 7 records

Client User Name	Client Host	Client Host Groups	Server Host	Server Host Groups	Duration	Application	Service Summary
adam	10.10.10.138	Corporate Wired User IP Pool	ad.mango.local (10.20.60.10)	Domain Controllers	00:00:16	NFS	smb (445/tcp)

- Maintain historical session table
- Correlate NetFlow to username
- Build User-centric reports

FTD Policies Based on ISE Context and Sec Groups



Security Groups
For Policy Export go to [Administration > System > Backup & Restore > Policy Export Page](#)

✎ Edit ✚ Add 📁 Import 📄 Export ✕ Delete ➡ Push

Icon	Name	SGT (Dec / Hex)	Description
☐	Auditors	221/00DD	Auditor Users Security Group
☐	BYOD	222/00DE	BYOD Device Security Group
☐	Contractors	223/00DF	Contractor Users Security Group
☐	Developers	224/00E0	Developer Users Security Group
☐	Development_Servers	301/012D	Development Servers Security Group
☐	Employees	235/00EB	Employee Security Group
☐	Guests	401/0191	Guest Users Security Group
☐	Network_Services	3/0003	Network Services Security Group

Add Rule

Name: ☒ Enabled Insert: **below rule** 4

Action: ☒ Allow ☐ Deny ☐ Log Only ☐ Log & Deny

Zones Networks VLAN Tags Users Applications Ports URLs **SGT/ISE Attributes** Logging Comments Inspection

Available Attributes Search by name or value

- Security Group Tag
- Device type
- Location IP

Available Metadata Search by name or value

- Auditors
- BYOD
- Contractors
- Developers
- Development_Servers
- Employees
- Guests
- Network_Services

Selected Source Metadata (0)

any

Add to Rule

Add a Location IP Address Add

Vulnerability-Aware Cisco Security

Using Vulnerability to Drive Threat Response in Firepower & ISE



- Allows the **import** of **Host** and **Vulnerability** Data



Host Profile

Domain:	Eshel \ Client_Backend \ Client_BDC
IP Address:	10.0.247.32
NutB00t Name:	
Device (Hops):	windows-domain client (0)
MAC Addresses (TTIs):	00:0C:29:03:0F:A2 (VMware, Inc.) (138) 00:0C:29:35:0F:5D (VMware, Inc.) (138) 00:11:22:33:44:55 (VMware, Inc.) (84) 0000:00
Host Type:	Host
Last Seen:	2018-01-08 05:29:49
Current User:	systeme user (LOCAL-BDC\user, LDAP)
View:	Contact, Explorer Connection, Events Intrusion, Events File, Events Network, Events

[Edit Host Status](#) [Mark All Resources](#)

Indications of Compromise (3) •

Operating System •

Vendor	Product	Version	Source
Microsoft	Windows	7	

Servers (5) •

Protocol	Port	Application Protocol	Vendor and Version
⊖ vsp	138	netbios-ssn	
⊖ tcp	139	pending	
⊖ tcp	3410	pending	
⊖ vsp	137	netbios-ns	
⊖ tcp	80	HTTP	Apache 2.2.3 (Debian)

Applications (11) •

Application Protocol	Client	Version	Web Application
⊖ pending	☐ DNS		
☐ HTTP	☒ Chrome	13.0.762.32	Web Browsing
☐ HTTP	☒ Chrome	24.0.1284.0	Web Browsing
☐ HTTP	☒ Chrome	27.0.1453.116	Web Browsing
☐ HTTP	☒ Chrome	36.0.1985.125	Web Browsing
☐ HTTP	☒ Chrome	4.0.265.1	Web Browsing
☐ HTTP	☒ Chrome	5.0.375.99	Web Browsing
☐ HTTP	☒ Internet Explorer	9.0.1	Web Browsing
☐ HTTP	☒ Internet Explorer	9.50	Web Browsing
☐ HTTP	☒ Safari		Web Browsing
⊖ HTTP	☒ Safari	4.0ppt	Web Browsing

User History •

Attributes •	Name
Host Criticality	Non-Critical
Cowd Cloud - Whitelist - Production	Non-Compliant

[Edit Attributes](#)

Host Profiles

White List Violations (17) •

Most Recent Malware Detections (100) •

Vulnerabilities (523) •

Name	Remote	Component	Port
ZeroLog, as used in Apple Safari before 5.0.6, allows remote attackers to execute arbitrary code or cause a denial of service (application crash) and application control via a crafted web site, a different vulnerability than other ZeroLog flaws listed in CVE.		Safari	
** RESERVED ** This candidate has been reserved by an organization or individual that will use it when announcing a new security position. When the candidate has been published, the details for this candidate will be provided.		Windows 7	
** RESERVED ** This candidate has been reserved by an organization or individual that will use it when announcing a new security position. When the candidate has been published, the details for this candidate will be provided.		Windows 7	
** RESERVED ** This candidate has been reserved by an organization or individual that will use it when announcing a new security position. When the candidate has been published, the details for this candidate will be provided.		Windows 7	

Vulnerability Names and IDs

Qualys – ISE Integration

ENDPOINT COUNT BY CVSS

Number of Endpoints Affected

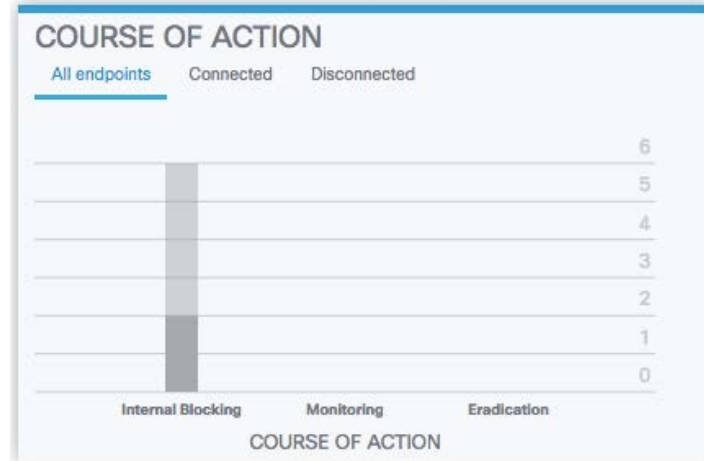
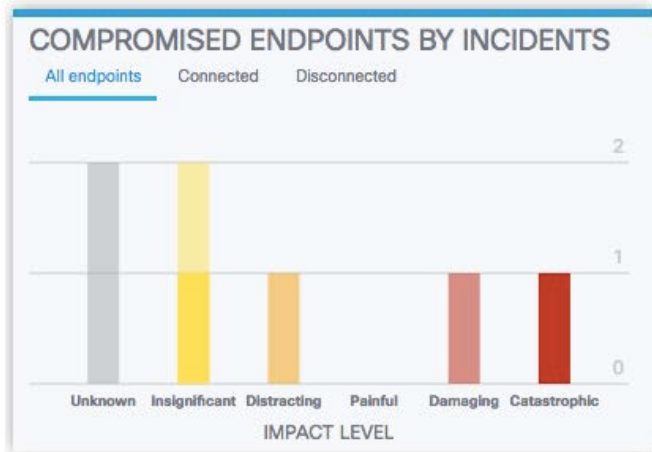


Rows/Page 10 1

Refresh	Add	Trash	Edit	ANC	Change Authorization	Clear Threats & Vulnerabilities	Export	Import	MDM Actions	Refresh MDM Partner Endpoint	Revoke Certificate
☐	MAC Address	Username	IPv4 Address	Vulnerability	Source	Score	Logical NAD L...	Connectivity	Hostname	Identity Group	Endpoint OS
☒	MAC Address	Username	IPv4 Address	Vulnerability	Source	Score	Logical NAD Loc	Connectivity	Hostname	Identity Group	Endpoint OS
☐	00:00:00:0F:42:40	test	0.15.66.64	QID-78027	Qualys	7.5	Location#All Loc...			Profiled	
				QID-124393	Qualys	6.8					
				QID-168127	Qualys	5.8					
				QID-123978	Qualys	5					
				QID-156740	Qualys	4					
				25 more vulnera...							

CTA/AMP – ISE Integration

Difference: vulnerable (Qualys) vs
compromised (CTA/AMP) endpoints



Quarantine

	MAC Address	Username	IPv4 Address	Threats	Source	Threat Severity	CoA	Confidence	Logical NAD Lo
×	MAC Address	Username	IPv4 Address	Threats	Source	Threat Severity	CoA	Confidence	
☐	00:00:00:00:00:01	dkb2	0.0.0.1	malicious host	CTA	Insignificant	Internal Blocking	Low	Location → All Loc
☐	00:00:00:00:00:02	dkb2	0.0.0.2	possibly unwanted app...	CTA	Unknown	Internal Blocking	High	Location → All Loc
☐	00:00:00:00:00:03	dkb2	0.0.0.3	malicious host	CTA	Insignificant	Internal Blocking	Medium	Location → All Loc
				malicious host	CTA	Insignificant	Internal Blocking	High	

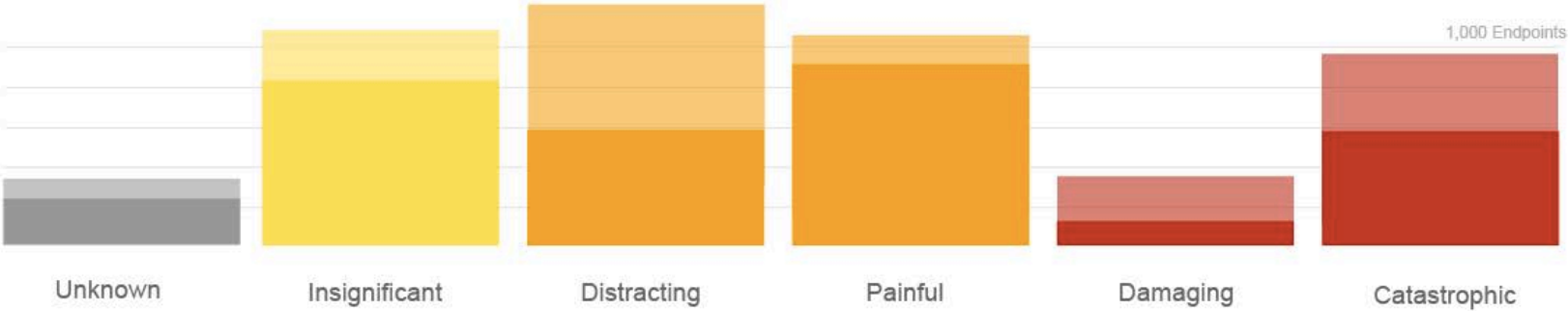
What is Threat Centric NAC: Threat

Threat EndPoints based on *Incidents and Indicators*

COMPROMISED ENDPOINTS

View Incidents

All endpoints | Connected | Disconnected



Rows/Page 50 1 / 12 561 Total Rows

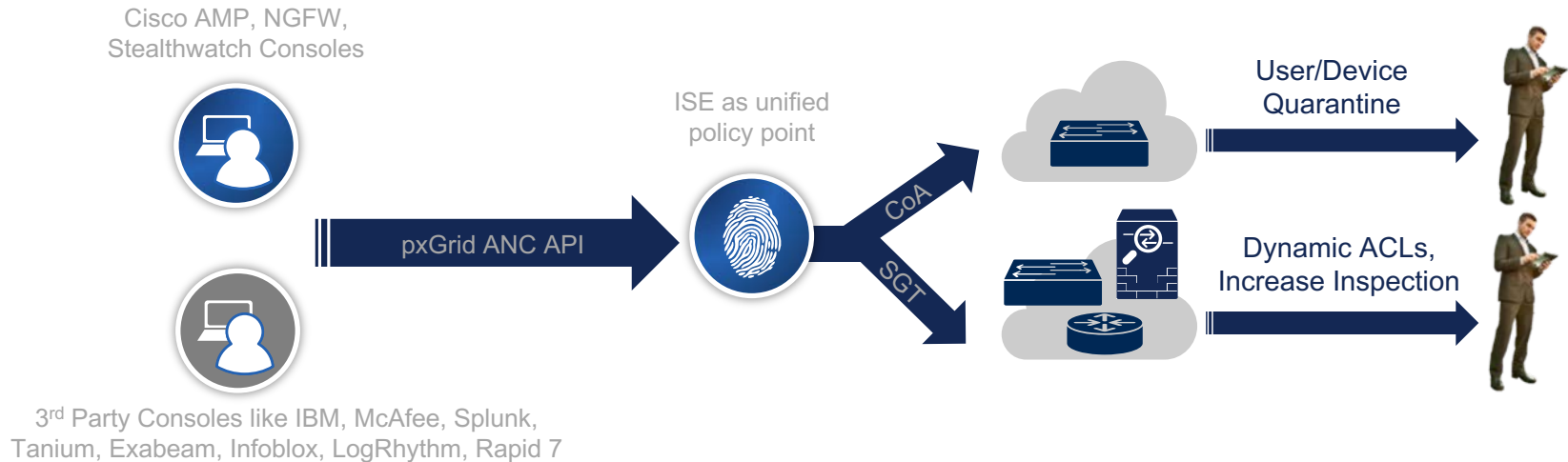
⌂ + 🗑️ 🔗 Change of Authorization ANC Device Registration MDM Actions Clear Threats & Vulnerabilities Import Export 🔍 ⚙️

☐	MAC Address	User	IP Address	Threats	Source	Threat Severity	Logical NAD Location	Connectivity	Hostname
✕									
☐	98:4B:4A:09:5A:28	Big_Boss	1.1.2.1	Microsoft Internet Explorer	AMP	Likely Impact: High	SJC 19/Floor 3	Connected	ASBHAT-M-20

Incident Response: Rapid Threat Containment

Cisco AMP, Firepower, Stealthwatch, ISE & CSTA Partners

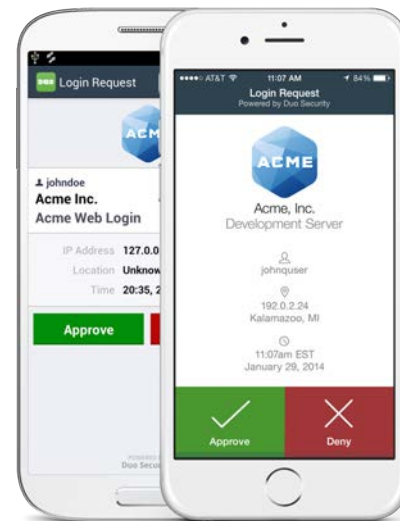
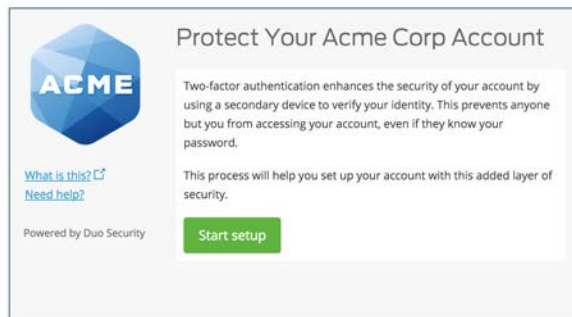
- “Rapid Threat Containment” – automatically or manually quarantine devices or spawn investigations
- Enlist other Cisco infrastructure in the network response – such as dynamic ACLs on switches and ASA or increase IPS inspection levels



DUO

Duo's Approach is Easy and Reduces Cost

- 1 Instantly integrates with all apps
- 2 Users self-enroll in minutes
- 3 Authenticate in seconds



Endpoint Remediation

×

Notify users when they are using out of date versions of:

☒ Browsers

☒ Flash

☒ Java

Anonymous Networks

×

For users that request access through proxies, Tor, or VPN:

Deny access

Trusted Devices

×

☐ Do not remember devices.

☒ Users may choose to remember their device for 30 days

Trusted Networks

×

☒ No trusted networks.

☐ Allow access without 2FA from these networks:

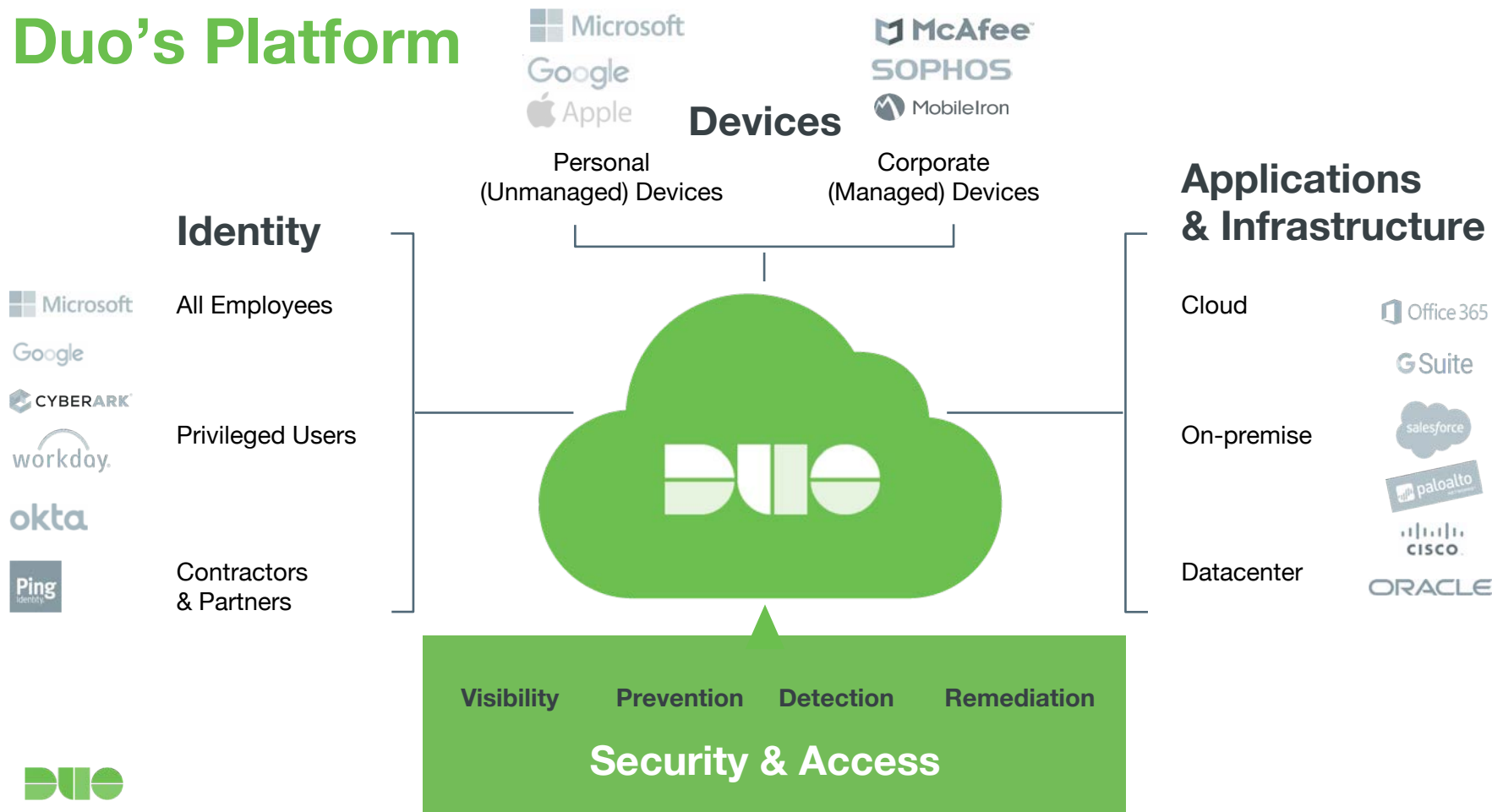
Ex: "192.0.2.8, 198.51.100.0-198.51.100.20, 203.0.113.0/24"

☐ Require enrollment from these networks.

3 Key Points About Duo's Security Policies

1. Centrally build policies for all apps
2. Web based policy management
3. Customize for user groups & apps

Duo's Platform



Flexible Authentication Options for your users

**Push, soft token, SMS,
Phone Call, U2F, Wearables,
Biometrics, HW Tokens**



Verify End User Devices

Allow only compliant devices to access work applications

1. Mobile (iOS and Android)

- a. Natively using Duo Mobile app. MDM alternative.
- b. Integration with MDM platforms.

1. Non Mobile (Windows, Mac, Linux, ChromeOS)

- a. Natively using browser data. No agents.
- b. Integration with endpoint management platform.

